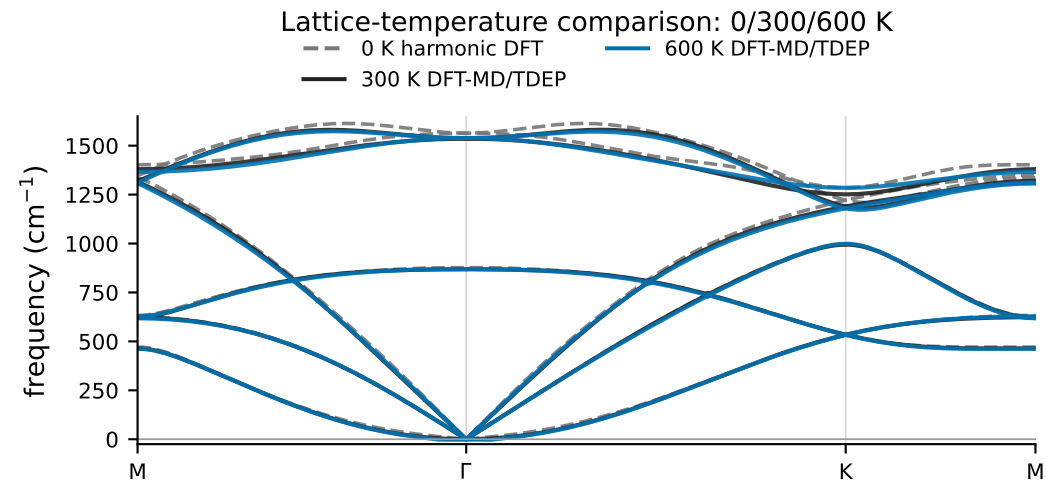
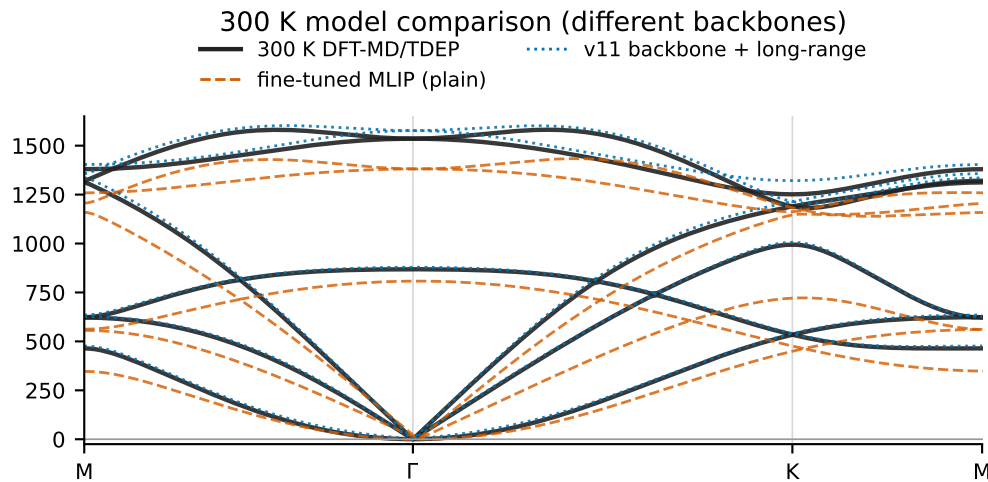


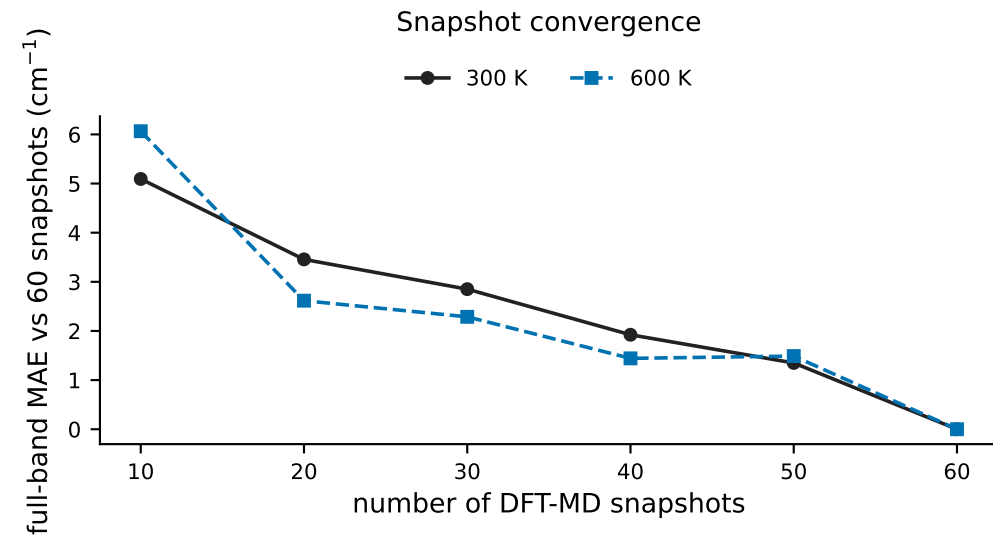
Graphene lattice-temperature phonons: DFT reference and MLIP comparison



0→300 / 300→600 K full-band MAE = 15.2 / 5.3 cm^{-1}
 Γ_{max} : 300 K 1536; 600 K 1539 cm^{-1}
 K_{max} : 300 K 1251; 600 K 1284 cm^{-1}



plain fine-tuned MLIP MAE = 95.9 cm^{-1}
 v11 + long-range MAE = 18.0 cm^{-1}
 K kink $|\Delta v|$: DFT 37.7, v11+LR 20.0



300 K: 50→60 MAE 1.35; K first/last Δ 41.7 cm^{-1}
 600 K: 50→60 MAE 1.49; K first/last Δ 12.2 cm^{-1}